Advanced AI/ML Topic Approval Form

The purpose of this document is to help you clearly explain your AI/ML topic, project scope, and timeline. Identify each of these areas so that you will have a complete and realistic overview of your project. Your instructor cannot sign off on your project topic without this information. Note: You must fill out and submit this form. Space beneath each number will expand as needed. Note: Any costs associated with developing the application will be the responsibility of the student.

INFORM INSTRUCTOR: Potential use of proprietary company information: (Y/N) **N**

PROBLEM AND GOAL: Explain the business problem or organizational need that you identified.

Subscription-based businesses often face high customer churn, which increases acquisition costs and reduces long-term profitability. Without predictive tools, companies struggle to identify at risk customers early enough to retain them through targeted interventions.

Identify one SMART goal of the proposed project using a SBI goal format.

Specific: Develop and evaluate an AI/ML churn prediction model.
Behavioral: The model will analyze customer usage and engagement data to produce a churn risk score for each active customer.
Impact: Achieve at least 80% AUC-ROC score on test data, enabling the business to reduce churn by 10% within 6 months of deployment

DATASET AND ANALYTICAL METHOD: Briefly describe the dataset you will be working with by discussing the format, size, and features. Please also cite the source(s).

- **Dataset**:
  A public customer churn dataset (CSV format) from the Kaggle "Telco Customer Churn" dataset will be used. It contains 7,043 customer records with features including customer tenure, contract type, monthly charges, payment method, service usage, and churn labels (Yes/No).

  Source: Kaggle. "Telco Customer Churn." *Www.kaggle.com*, 2017, www.kaggle.com/datasets/blastchar/telco-customer-churn.

Identify the AI/ML analytical method you plan to use.

- **Analytical Method**:
  A binary classification approach will be applied. Candidate models include Logistic

Regression, Random Forest, and XGBoost. Feature importance and SHAP values will be used for interpretability. Evaluation will use AUC-ROC, precision/recall, and lift metrics.

PROJECT MANAGEMENT AND CHARACTERISTICS Outline major tasks and the schedule of the proposed project in a simple table. This will include anticipated timelines, due dates, costs, etc.

Task	Start Date	End Date	Deliverable
Problem definition & scoping	09/01/25	09/05/25	Documented problem & goal
Data collection & preprocessing	09/06/25	09/15/25	Clean dataset
Model training & tuning	09/16/25	09/30/25	Trained models
Evaluation & performance analysis	10/01/25	10/10/25	Metrics report
Final documentation & approval	10/11/25	10/15/25	Completed project report

Identify the functional and nonfunctional requirements of your project.

- **Functional Requirements**:
- Accept structured customer data as input.
- Output a churn risk score per customer.
- **Nonfunctional Requirements**:
- Model accuracy ≥ 80% AUC-ROC.
- Predictions generated in < 2 seconds per customer.
- Scalable to at least 50,000 customer records.

Describe any risks to the project and identify mitigation strategies.

- **Risks & Mitigation**:
    - *Risk*: Dataset imbalance (few churners).
      Mitigation: Use resampling (SMOTE) and class weights.
    - *Risk*: Model overfitting.
      Mitigation: Use cross-validation and regularization.

    - *Risk*: Data Quality issues
      Mitigation: Perform thorough data cleaning, validate inputs, and establish clear preprocessing steps.

DESIGN AND DEVELOPMENT Identify the programming/development language(s) you will use.

Programming/Development Language(s): Python (Scikit-learn, XGBoost, Pandas, TensorFlow).

Describe the programming environment (integrated development environment)— including both the software and hardware—you will use to create the application.

Programming Environment: Jupyter Notebook (Anaconda distribution) with access to GPU-enabled cloud (Google Colab).

Identify the operating system(s) or platform(s) you will use to complete the project.

Operating System / Platform: Windows 11 for local testing; Linux-based Google Colab for model training.

This project does not involve human subject research and is exempt from WGU IRB review.

STUDENT'S SIGNATURE

Michael Williams

By signing and submitting this form, you acknowledge that you (the student) are responsible for any costs associated with developing and executing the application.

INSTRUCTOR'S SIGNATURE:

INSTRUCTOR APPROVAL DATE: